Queensland University of Technology

**EGB320 – Mechatronics Design II**

# Final Technical Report

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# Contents

# 1 Problem Description

The Mechatronics Design II project aimed to develop a *TRL 3 prototype* autonomous robot capable of completing a restocking task in a simulated grocery store environment. The robot was required to identify, collect, and restock items on shelving units autonomously, while demonstrating a robust design that could be applied in the future by *Colesworth Pty Ltd*. Each team member was responsible for the research, design, and testing for a specific subsystem that contributed to the overall system.

These included navigation, mobility, vision, and item collection. This report will focus on the **item collection subsystem**.

## 1.1 Subsystem Overview

The **item collection subsystem** is divided into two main categories: lifting and grabbing. The lifting mechanism purpose is to raise and lower a platform, enabling the robot to reach the object in the ground and then reach three different shelf heights. The grabbing mechanism is mounted in the raising platform and is designed to grasp an object to pick it up and then release it at the desired shelf.

For the complete product, both mechanisms work together to reliably being able to pick and place items of different shapes and sizes into their intended storage heights.

## 1.2 Specifications

Taking into account the overall project objectives, environmental constraints and design limitations given by *Colesworth Pty Ltd*, the functional requirements for the lifting mechanism were summarized in Table 1.

Table 1: Specifications for the lifting mechanism subsystem

| **Parameter** | **Value** | **Purpose** |
|---|---|---|
| Vertical range of motion | 250 mm | To reach ground and three shelf levels |
| Payload capacity | 0.5 kg | To carry items and gripper mech. |
| Angular deviation (Tilt) | $< 7°$ | Maintain stable item orientation |
| Speed | $\leq 3$ s full lift | Enable efficient task completion |
| Total subsystem mass | $< 500$ g | Minimize tilt and motor torque |
| Power supply | 6 V, $< 1.5$A | Compatible with main system power bus |
| Size | $< 200\,mm^3$ | Fit comfortable in aisles |

# 2 Background

## 2.1 Related Work

Vertical actuation systems and Gripper mechanisms are widely used in the robotic industry for warehouse automation, manufacturing/assembling, or packaging. Common approaches for lifting include telescopic actuators, screw driven lifts, scissor lifts, or belt drive systems. And for gripping robots often use parallel grippers, vacuum grippers, or pneumatic grippers.

When building prototypes, choosing materials of low cost and simple methods can accelerate the rate at which designs are made and tested, ultimately making better designs. It is important to balance ease of manufacturability with high reliability in order to make systems that can represent real life system's in a controlled environment. Given these constraints, similar projects often adopt linear actuators, four bar linkages, and scissor lifts due to their intuitive operation and ability to produce stable vertical motion.

## 2.2 Selection and Comparison

Several mechanical configurations and actuation methods were considered during the design phase. The selection process aimed to balance manufacturability, reliability, and vertical stability, while maintaining motion precision and low cost.

### 2.2.1 Comparison of Four-Bar Configurations

Three main variants of linkage geometries were evaluated in Table 2: the parallel four-bar linkage, straight-line four-bar linkage (Scott Russell), and scissor lift mechanism.

Table 2: Weighted comparison of lifting mechanism configurations

| **Configuration** | **Motion Type (×3)** | **Complexity (×1)** | **Stability (×3)** | **Cost (×1)** | **Total Score (/40)** |
|---|---|---|---|---|---|
| Parallel Four-Bar Linkage | 3 (9) | 5 (5) | 4 (12) | 5 (5) | 31 |
| Straight-Line Linkage (Watt/Chebyshev) | 4 (12) | 3 (3) | 3 (9) | 4 (4) | 28 |
| Scissor Lift Mechanism | 5 (15) | 2 (2) | 5 (15) | 0 (0) | **32** |

The scissor lift mechanism configuration was selected mainly because of its motion type. Unlike the parallel four-bar linkage whose motion is arc-like, the scissor lift mechanism raises vertically. This was a major influence for the team as having true linear motion simplified the software calculations by allowing the robot to be at the same distance to the shelf regardless of the wanted height. Although it has the highest complexity among the three configurations, it also has higher stability. This is also of great importance considering the robot is mobile and has obstacles such as ramps. Having the center of gravity in the center of the robot allows greater speed and more control over the different angled ramps. The decision was also influenced after some comparison testing using 3D printed linkages, shown in Figure 1 below:


Figure 1: Testing the lifting options motion and stability.

#### 2.2.2 Gripper Type Selection

The gripper selected for this subsystem was a simple parallel gripper driven by a rack-and-pinion mechanism. This configuration was chosen due to its mechanical simplicity, reliability, and ease of integration with the existing lifting structure.

The parallel motion of the grippers ensured even gripping forces on objects of varying shapes, suited for handling packaged items during restocking. Additionally, the rack-and-pinion allowed conversion of rotational motion from the motor into linear movement, minimizing complexity while maintaining consistent performance.

#### 2.2.3 Actuation Method Comparison

In addition to the linkages, three actuator types were evaluated for both lifting and gripping: DC motors with and without encoders and servo motors. Table 3 presents the weighted comparison.

Table 3: Comparison of actuator types for lifting and gripping

| **Actuator Type** | **Precision (×3)** | **Torque (×2)** | **Speed (×2)** | **Cost (×1)** | **Total Score (/40)** |
|---|---|---|---|---|---|
| DC Motor | 2 (6) | 4 (8) | 5 (10) | 5 (5) | 29 |
| Servo Motor | 5 (15) | 3 (6) | 3 (6) | 4 (4) | 31 |
| DC Motor with Encoder | 4 (12) | 4 (8) | 5 (10) | 4 (4) | **34** |

DC motors with encoders were ultimately chosen for both the lifting and gripping mechanisms. For the lifting mechanism, the encoded DC motor provided the required continuous rotation and sufficient torque to drive the scissor mechanism across its full 25 cm range. For the gripping mechanism, a plain DC motor and limit switches was planned and tested at first, but later changed for the same encoded motor as the lifting mechanism after finding a cost-effective option that maintained accurate position feedback while being cheaper than the other options.

# 3 Design

## 3.1 Overall System Design

The item collection subsystem is positioned in the center of the chassis and connects with it through the bottom base. Even though they were designed separately at first, the design eventually merged through the bottom base. Its symmetrical and centered arrangement allowed the whole robot to maintain balance and stability during its operation.

Figure 2: Complete System Design

As it can be seen in Figure 1 above, it consists of a vertical scissor-lift mechanism driven by a geared DC motor with encoder feedback and a lead screw. This allowed smooth and controlled elevation of the gripping system. At the top of the lift, a belt driven rack and pinion gripper was mounted to securely grasp and release items of varying widths.

## 3.2 Mechanical Design

The design evolved through multiple iterations in CAD, optimizing the linkage geometry, spacing, clearances and tolerances, and material thickness to achieve a functioning product that can have 25 cm of vertical travel with minimal lateral deviation.

### 3.2.1 Lifting Mechanics

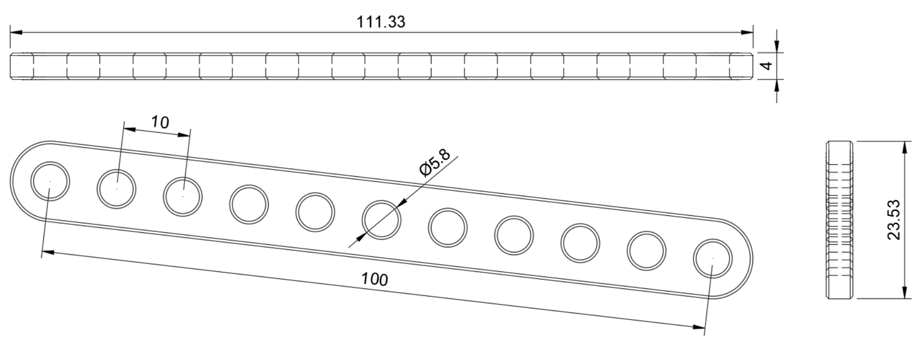



Figure 3: 2D Drawing of linkages(in mm).

Figure 3 above shows a CAD model of the linkages and the basics of the scissor lift mechanism functionality. Each linkage is 110$mm$ long and has 11 holes that fit a custom pivotal screw. Connecting two linkages by the middle hole forms a cross, and by having three repeated crosses on top of each other the scissor is able to reach a maximum height of around 270$mm$. Taking into account it is not the final version used, the initial CAD is seen below in Figure 4. The model illustrates how the three height scissor formation is mounted each side of the bottom and top base through the same pivotal screws. One of these connections between the scissor and the base only allows for rotational movement, while the other can also translate around the x-axis; hence, allowing the scissor to extend vertically.

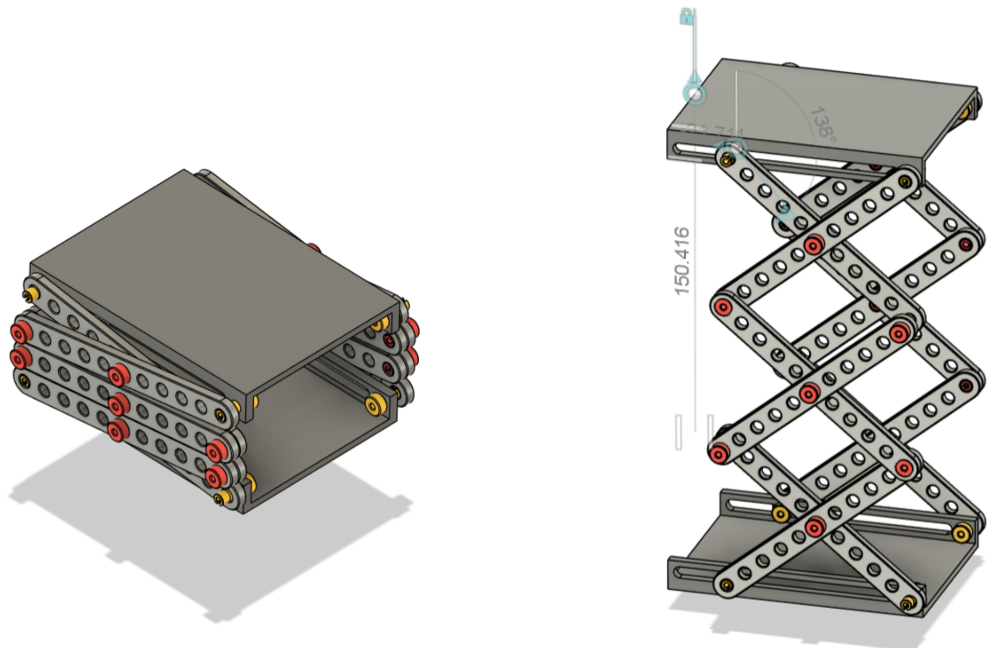



Figure 4: Isometric View - Closed and Half Open Scissor Lift layout

A lead screw mechanism is used to convert the rotational motion of the actuator into linear displacement for the scissor lift. The selected *20N Geared DC Motor with Encoder* provides sufficient torque and precise position feedback for controlled lifting. The motor shaft is connected to the lead screw using a 3D-printed shaft coupler [8] and the lead screw is mounted through a ball bearing [3] on the other side (allowing for free rotation). Hence, as the motor rotates, the lead screw does as well at the same speed, which then moves the lead nut [6] linearly along the screw. This nut is attached to a secondary coupler [5] that links to the pivot screws in the base. , effectively translating rotational motion into a stable vertical lift. Having the lead screw increases the mechanical advantage of the system by behaving like a gearbox, given its lead the screw trades speed for force. Moreover, the added friction acts as a lock that prevents the scissor lift to fall without the need to add continuous power. This approach minimizes mechanical stress while maintaining accuracy and repeatability for multiple tries.

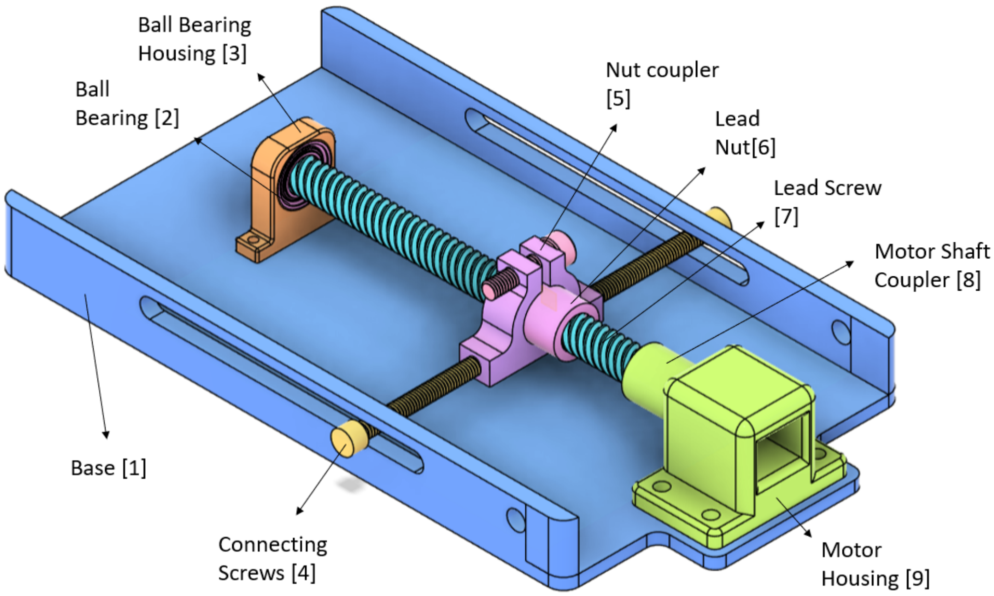



Figure 5: CAD model of the lead screw assembly

### 3.2.2 Gripping Mechanics

The gripping mechanism operates using a belt driven rack-and-pinion system powered by a DC motor with an encoder. Figure 6 illustrates the gripping mechanism exploded into two parts. Looking at the first half in the back, it can be seen that the motor's rotational motion is first transmitted through a timing belt [2][4] to a small gear [9]. This design was chosen because the motor should be located at the top of the base while its rotation should be kept lower. The former to keep the center of mass balanced, and hence reduce the tilt and lifting force required. And the latter lets the grippers be located further down, which is necessary to be able to pick the object from ground level.

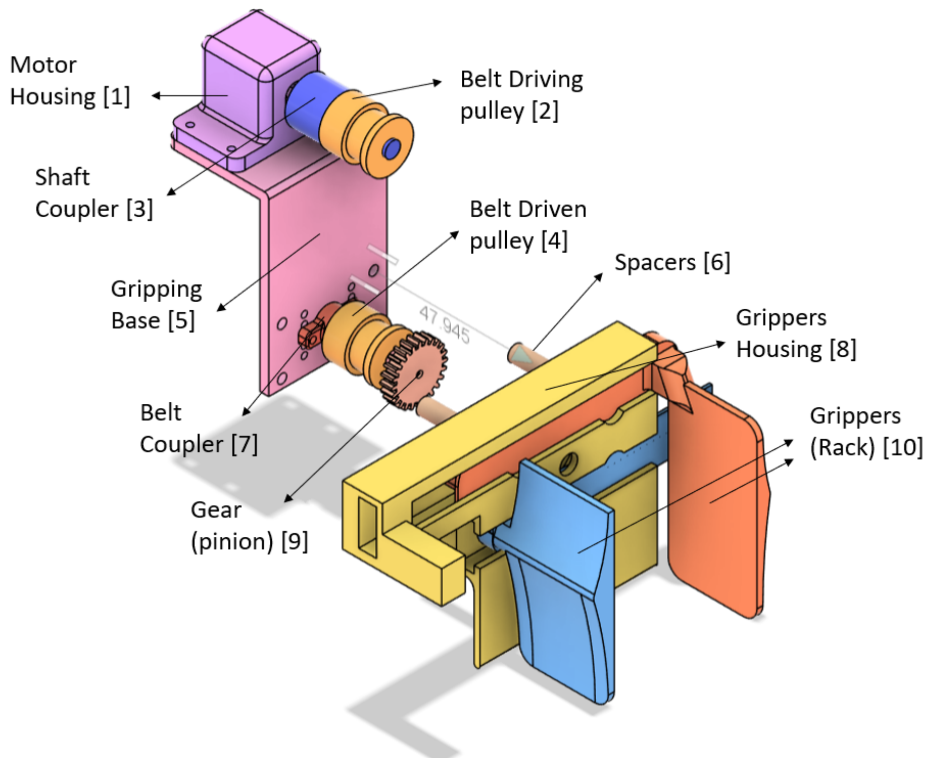
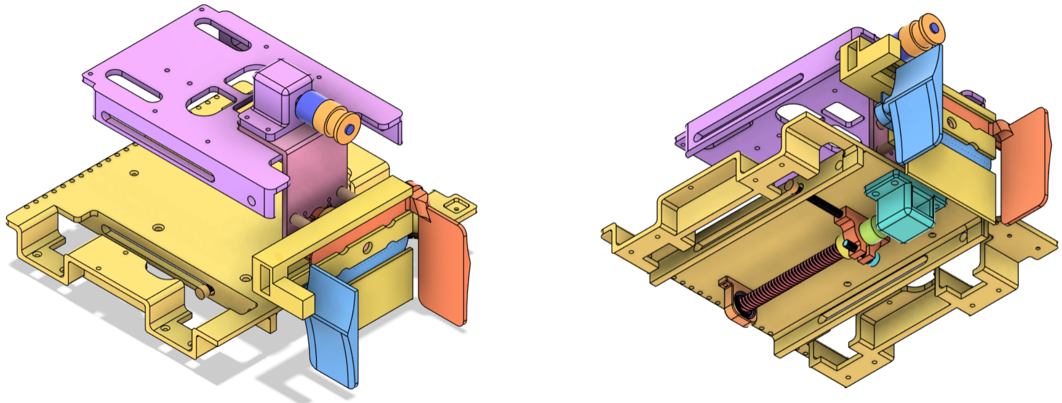



Figure 6: CAD model of the rack-and-pinion parallel grippers

The first half described then connects to the grippers through some spacers [6]. In the front it is evident that as the gear rotates, it moves the rack linearly, causing both grippers [10] to move symmetrically and open or close around the object. The encoder provides precise feedback for controlled gripping force and position, enabling reliable and repeatable motion suitable for gripping items of varying sizes during restocking tasks.

#### 3.2.3 Final Version

All components of the lifting and gripping subsystem were modeled in *Autodesk Fusion 360* and then 3D printed. Several design iterations were produced throughout this prototyping phase to improve tolerances, alignment, and try to optimize the fit while keeping friction low. Each version was evaluated for mechanical stability and smooth operation and was improved to get to the final version shown below.



(a) Top View (b) Bottom View.

Figure 7: Final Item Collection subsystem CAD assembly, isometric views.

Figure 7 illustrates the final version of the Item Collection subsystem, integrating both the lifting and gripping mechanisms. The lead screw assembly is positioned at the base of the chassis, while the gripping mechanism is mounted on the top base, as shown in the section analysis in Figure 7b. Note that the scissor linkages are omitted due to computational constraints, however they can be seen in Figure 4 and in the Appendix A. The final prototype achieved lifting motion with accurate vertical positioning to align with shelf heights while maximizing space for the electrical and mobility components. Placing the lead screw at the bottom allowed the main computer (Raspberry Pi 5) to be mounted on the chassis, while the front-mounted gripping system provided room at the top for the power system. The full robot design is provided later.

## 3.3 Electronics and Software Design

The robot's power and control system is organized to ensure efficient energy distribution and reliable communication between subsystems. Figure 8 below shows the flow chart of the power system. A 7.4 V battery is mounted on the top base of the robot along with a voltage regulator. This power module is able to distribute both regulated (at 5.1V) and unregulated (VBAT) outputs. The regulated voltage powers the Raspberry Pi 5 via a USB-C connection, while the unregulated line supplies the quad motor controller, which drives the four DC motors (two for mobility and two for item collection). The exact distribution of the electrical components can be seen in Appendix A.

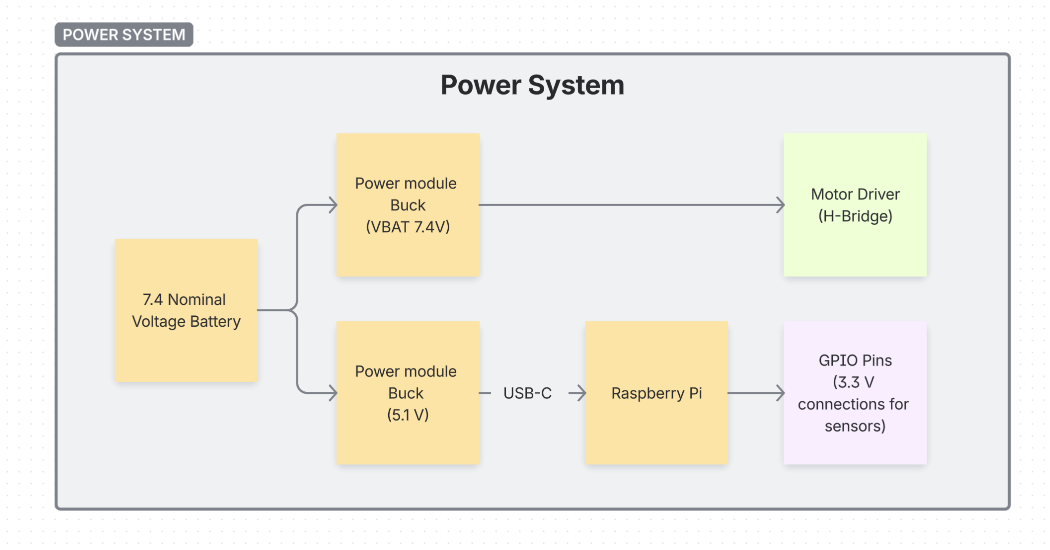



Figure 8: Power configuration diagram

The software and control communication is established through a serial interface between the Raspberry Pi and the motor controller. The navigation subsystem tells the motor controller when to open/close the grippers or lift to a certain level by sending a Serial Command. The command sent tells the motor controller which of the four motors to move, at which speed, whether to use positional or velocity control for the encoders, and direction of travel. The controller then reads the serial and is responsible for executing the commanded actions by generating PWM signals to drive the motors to regulate their motion. It also reads feedback from the encoders to monitor speed and position, ensuring accurate and stable operation.

When the system is turned on, the motor controller reads the position of the grippers and lead screw and marks them as starting positions. Whenever navigation requires the functions, the function *'liftlevel(level)'* creates a serial message such as $'l0'$ which is read by the motor controller and uses positional control taking ground level as 0, to raise the lift to the three different levels. Moreover, the function *grip(job)*, writes a serial message such as *mo* (Open) which the motor controller interprets and moves the grippers to their initial position, (open). For the closing position, the controller reads the encoder values, grabbing the object when the velocity reaches certain threshold. See Figure 9 below:

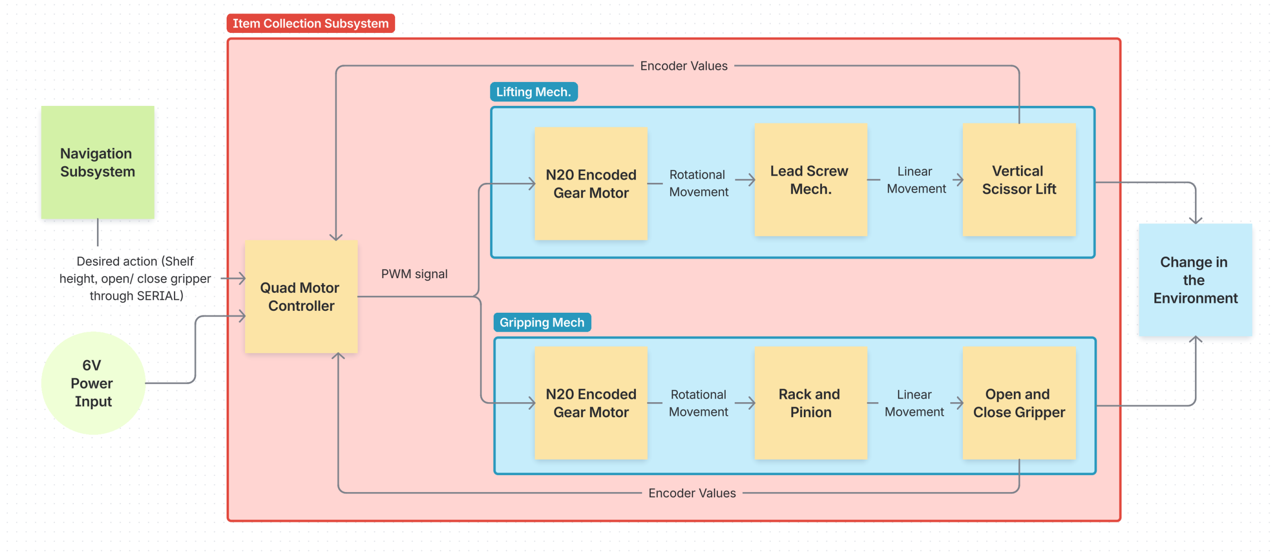



Figure 9: Item Collection logic flow diagram

### 3.4 Subsystem Integration

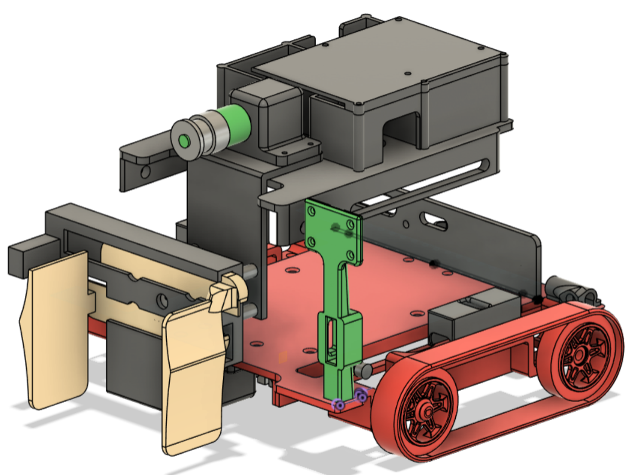


Figure 10: Final CAD model - Physical Integration

As shown in Figure 2, the final CAD model illustrates how the four main subsystems are integrated into a cohesive design. Physically, the mobility subsystem provides the chassis where all of the components (including the whole item collection subsystem) are mounted. This include some of the power components such as the Raspberry Pi and Motor controller, or the Camera and tank treads. The mobility system is also in charge of controlling the tank treads through two motors and the motor controller, which provide precise control of the robot's velocity, position, and turning angle. This configuration allows for turning on its own spot, an important feature given the constraint space. Additionally, the camera is mounted strategically so that the vision subsystem can perceive the environment and have a good field of view that allows the robot to detect items, markers, distances, colors even while moving. The three subsystems are then all applied through navigation so that the robot can autonomously move around the environment, detect objects and pick and place them in the desired shelves.

# 4 Design Validation

## 4.1 Testing Procedures and Results

A series of experimental tests were conducted to verify that the item collection subsystem met its design specifications, stated in Section 1.2. In order to have a working robot, aspects like lifting height, speed, load capability, and gripper performance was evaluated.

### 4.1.1 Range of Motion and Angular Deviation

The purpose of this test was to determine the total vertical travel distance of the lifting mechanism and to measure any angular deviation or tilt occurring during operation. The subsystem was positioned on a flat surface, and the scissor lift was lifted from its lowest to highest position while recording the height and tilt. The height was measured from the flat surface to the bottom of the gripper, given this is the height at which the object is picked. Images of this were captured to visually demonstrate the motion range, shown in Appendix C: Figure 14.

The results (Table 4) confirmed that the lift achieves the required height range. And that the angular deviation starts to become noticeable from the level 3. Even though this significantly reduces stability, the robot does not need to move more than 100mm when the lifting mechanism is at level 3. The tilt amount is acceptable for this controlled conditions, although they could become an issue, especially for less controlled environments. Moreover, the mechanism proved to be able to reach the three shelf heights, and surpass the vertical range of motion set in the specifications as 250mm

Table 4: Range of motion and angular deviation test results

| Test | Height (mm) | Tilt (°) | Pass/Fail |
|---|---|---|---|
| Ground Level | 10 | 0 | Pass |
| Level 1 | 75.1 | 0 | Pass |
| Level 2 | 165 | 1 | Pass |
| Level 3 | 237 | 4 | Fail Tilt |
| Max height | 254 | 5 | Fail Tilt |

#### 4.1.2 Speed and Payload Capacity

The next test aimed to evaluate how different payloads affected the lifting speed and the ability of the motor/lead screw system to reach the full extension height. The payload weight was taken only considering additional weight, not taking into account the current 350gr the top base weighs. The extra weight was applied incrementally while the time for the platform to reach the highest position was recorded. Each trial was repeated three times to obtain an average speed.

Table 5: Lifting speed and payload capacity test results

| Payload (g) | Avg Lift Time (s) | Maximum Height Reached (mm) | Pass/Fail |
|---|---|---|---|
| 0 | 6.13 | 237 | Pass |
| 50 | 6.87 | 235 | Pass |
| 100 | 8.04 | 235 | Pass |
| 150 | 10.15 | 232 | Pass |
| 200 | N/A | N/A | Fail |

The results demonstrated that the subsystem maintained consistent performance for typical payloads expected in the restocking task. At higher loads, the lifting speed decreased slightly, and at higher payload weights, the scissor lift mechanism was not able to lift it. Finally, the required speed according to the initial specifications was not met for neither weight, taking more than double the desired time ($> 3s$).

#### 4.1.3 Object Variety

To assess the gripper's adaptability, tests were performed using different objects in size, shape, and weight. These were lifted from level 0 to level 3 and back to level 0 to match the project's conditions. The objective was to confirm the gripper's ability to securely grasp and release objects without slippage or misalignment. Refer to Figure 15 in Appendix C for examples of test objects.

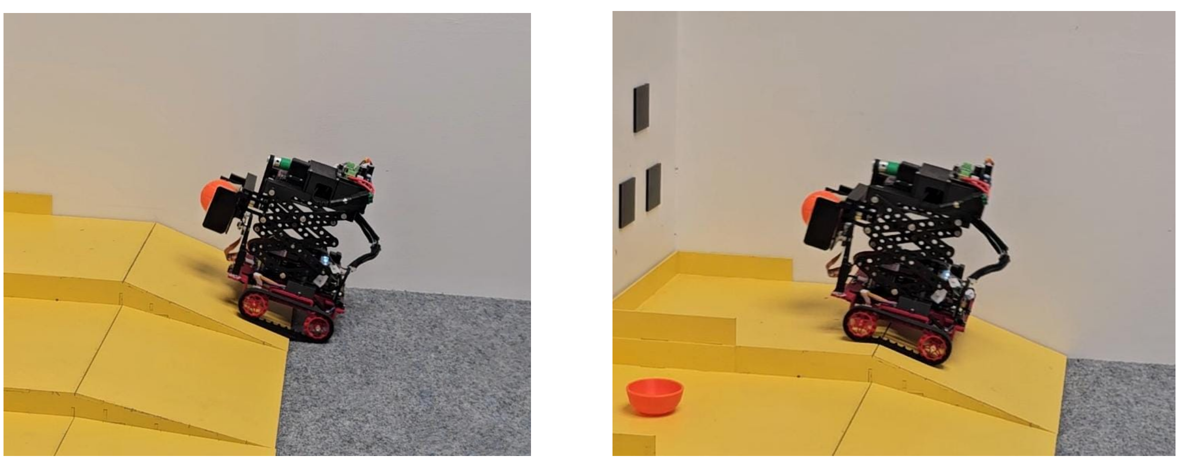
Table 6: Gripper performance with different object types

| **Object** | **Bottle** | **Bowl** | **Mug** | **Weet Bolts** | **Cube** | **Ball** | **Bowl + 50g** | **Bowl + 100g** |
|---|---|---|---|---|---|---|---|---|
| **Mass (g)** | 6 | 12 | 16 | 19 | 21 | 25 | 62 | 112 |
| **Pass/Fail** | Pass | Pass | Pass | Pass | Pass | Pass | Pass | Fail |

The results indicated reliable gripping for most objects tested. Some lightweight or irregularly shaped items showed some misalignment, which could be improved through finer motor control or adding compliant grip surfaces. However the objects did not fall, even when the robot was moving; and having passed the bowl + 50 gr test indicate a safety factor of 2.

#### 4.1.4 Integration Testing

Finally, integration tests were conducted to validate that the subsystem operates correctly when mounted on the robot. The robot was lifted to level 1 while carrying an item, and then moved up and down the steepest ramp, including stops. The subsystem maintained stability and did not drop the payload during movement, although it did move the item slightly. Moreover, this was then repeated with level 2 and it was evident that the robot looses its center of balance when having the item collection subsystem higher than level 1.



(a) Ramp Beggining (b) Ramp End.

Figure 11: Ramp testing at Level 1.

# 5 Conclusion

Based on the design validation results, the developed item collection subsystem successfully met the majority of the specifications outlined in Section 1.2. The scissor-lift mechanism achieved the required vertical range of 250 mm, demonstrating smooth and consistent motion with minimal lateral deviation for the first two levels, and a higher, yet acceptable tilt for the maximum height (under $5^\circ$ tilt). The combination of the lead screw and geared DC motor provided sufficient torque to lift the 0.35 kg top base, although the speed was compromised and increased the raising time above the $3s$. As for power, the mechanical advantage introduced by the lead screw allowed the platform to hold its position without continuous power which improved energy efficiency.

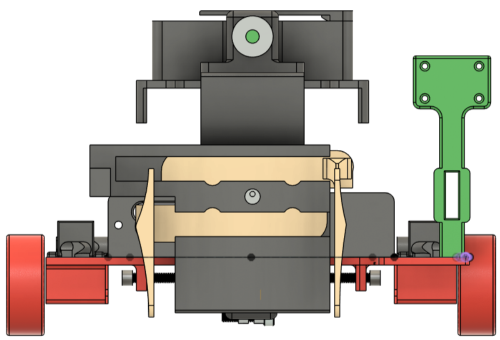
The gripping subsystem also performed reliably during testing, successfully grasping and releasing objects of various shapes and sizes. The encoder allowed for consistent opening and closing positions, while maintaining good grip force.

Despite its success, several areas for improvement were identified; having as a main concern the durability and reliability over extended periods of time. The scissor lift mechanism needs a balance between being rigid, for stability, without introducing to much friction that could make the motors stall. Therefore, replacing 3D-printed joints with metal or ball-bearing pivots would improve precision and durability over extended use. The scissor lift linkages could also be laser cut into thick acrylic that provides more structural rigidity and less friction than plastic. Finally, The scissor lift's speed could be increased by optimizing the lead screw pitch or using a higher-speed motor without compromising torque.

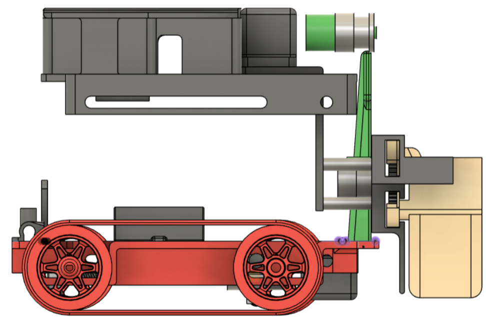
In conclusion, the final subsystem met its functional and design requirements, in order to provide a lifting and gripping solution for autonomous restocking. The results demonstrate that the design meets a TRL 3 prototype, capable of being refined into a more robust system with better materials for similar applications in future iterations.

# Appendix

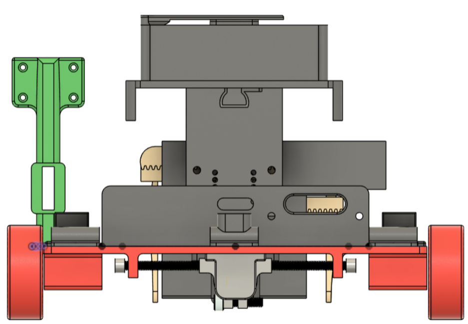
## Appendix A: CAD, and Robot images



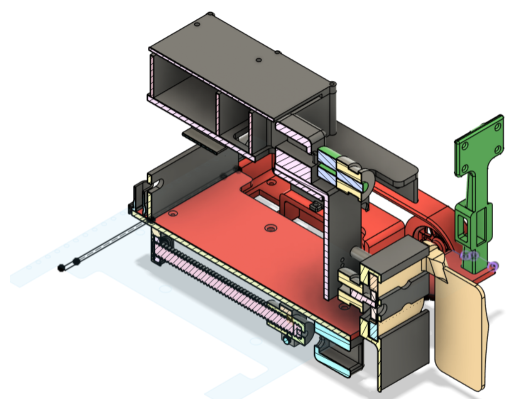
(a) Front view.



(b) Side view.



(c) Back view.



(d) Isometric cut view.

Figure 12: Final CAD rendering.

## Appendix B: Code Movement functions

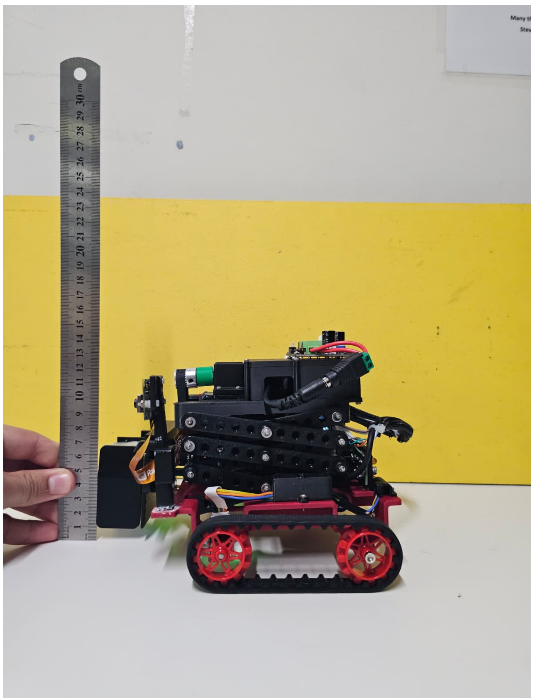
```python
import serial
import time

# Initialize serial communication
ser = serial.Serial(port='/dev/ttyACM0', baudrate=9600, timeout=1, stopbits=serial.STOPBITS_ONE)
```

```python
# ser = serial.Serial(port='COM4', baudrate=9600, timeout=1,
    ↪ stopbits=serial.STOPBITS_ONE)

# Define lift levels
LEVEL_0 = 0
LEVEL_1 = 1
LEVEL_2 = 2
LEVEL_3 = 3
LEVELS = [LEVEL_0, LEVEL_1, LEVEL_2, LEVEL_3]

def lift_level(level):
    """Moves the lift to the specified level."""
    message_1 = 'l' + str(level) + '\n'
    ser.write(message_1.encode())

def grip(job):
    """Controls the gripper mechanism.
       job = 'o'     open
       job = 'm'     close
       job = 's'     stop
    """
    if job == 'o':
        ser.write("mo\n".encode())
    elif job == 'm':
        ser.write("mm\n".encode())
    elif job == 's':
        ser.write("ms\n".encode())
```

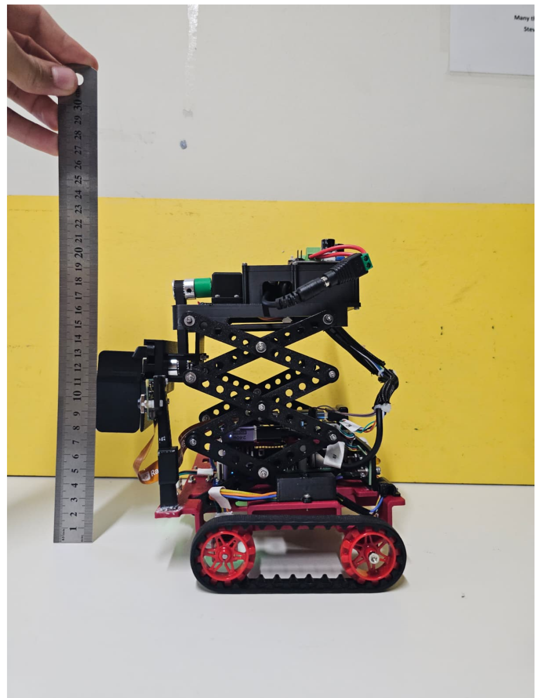
Listing 1: Item collection code

# Appendix C: Testing Images

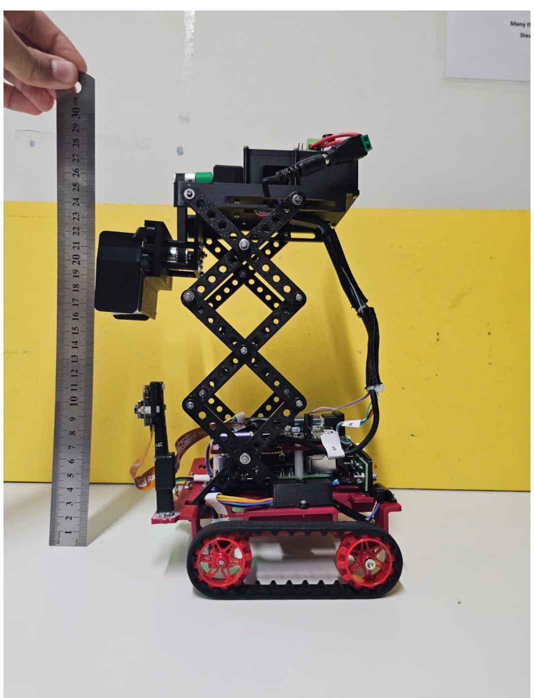
### 5.0.1 Range of motion tests



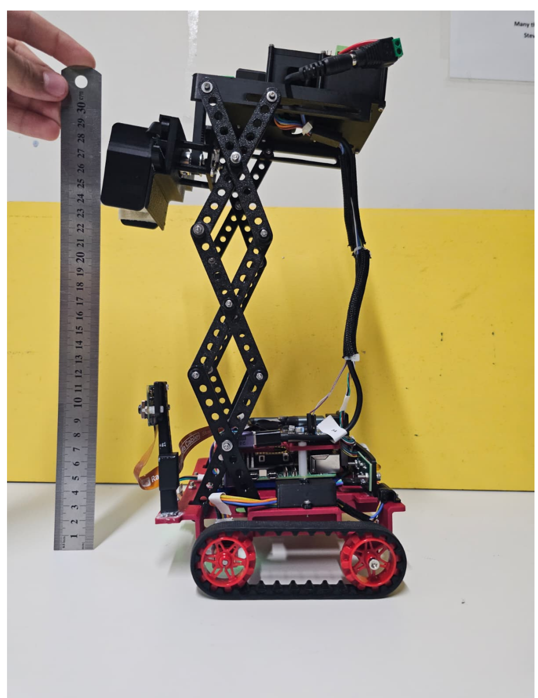
(a) Range of motion L0.



(b) Range of motion L1.



(c) Range of motion L2.



(d) Range of motion L3.

Figure 13: Range of motion tests at levels L0–L3.

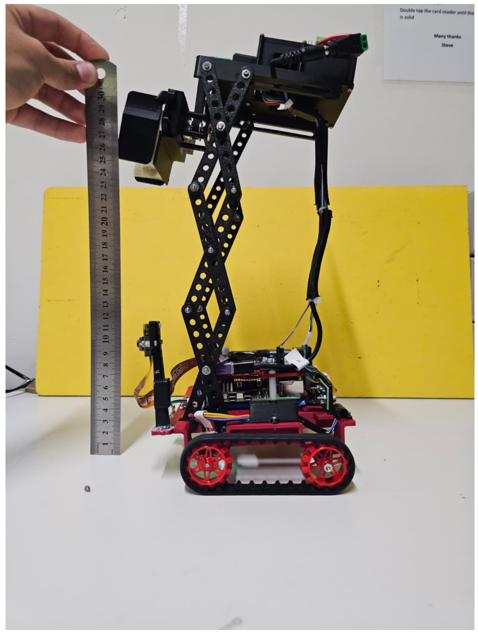


Figure 14: Range of motion Lmax.

#### 5.0.2 Object Variety tests

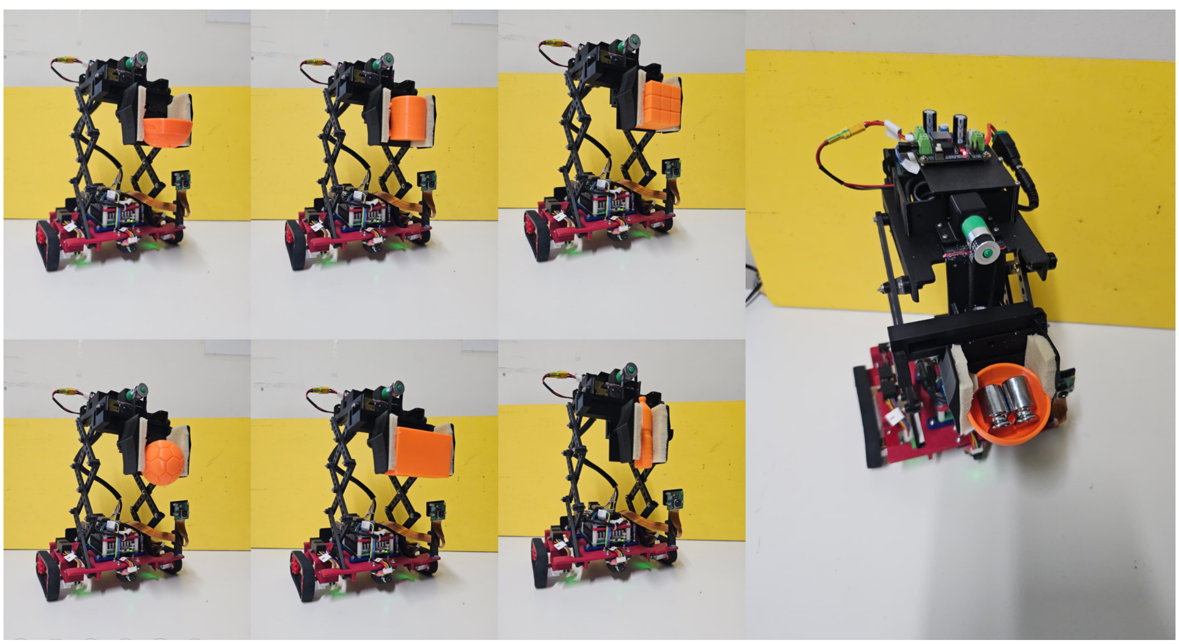


Figure 15: Robot picking up different objects